

Cloud-Robust Algal Bloom Early Warning for Johor Strait Fish Farms

BUCKET 8 - COASTAL AND AQUACULTURE

For the ~110 licensed sea-based fish farms in the Johor Strait (Singapore Food Agency, SFA) and their Johor-side counterparts, who lost 160 tonnes of fish to harmful-algal-bloom (HAB) fish-kill events in a single 2014-2015 episode, this project fuses Sentinel-2, Sentinel-3 OLCI and Sentinel-1 SAR with a partial-observation deep model trained to keep forecasting skill through the region's persistent tropical cloud cover, scored on lead time before a fish-kill event rather than per-pixel chlorophyll accuracy.

VIABLE

The idea passes all six hard constraints but carries a serious, verified label-scarcity risk (no continuous Johor Strait/Singapore bloom or fish-kill record exists past ~2016) and a narrower-than-advertised novelty margin against two very recent 2025-2026 papers that must be explicitly benchmarked against, not assumed away.

Confidence: Medium. Confidence would rise to High with: (a) a week-1 manual press-archive trawl confirming enough usable fish-kill event-dates to make the lead-time metric statistically meaningful, and (b) a from-Malaysia test of Copernicus Data Space Ecosystem STAC/COG cropped access to confirm dataset volumes stay under budget. It would fall to Low if that trawl turns up fewer than ~5 usable events, since the entire evaluation protocol depends on having a real, if small, ground-truth event set.

Hard constraint check

Constraint	Verdict	Reasoning
H1 • Novelty	CAVEAT	genuine delta over two close 2025-2026 papers via SAR-anchored partial-observation fusion + lead-time-to-fish-kill objective + tropical cloud-persistent regime, but the margin is narrower than the pitch implies and must be defended explicitly against arXiv:2606.17242 and arXiv:2510.02763.
H2 • Not generic	PASS	not any of the 8 banned patterns; specific sensor fusion + lead-time framing is not a generic 'apply model to region' project.
H3 • Imagery is primary input	PASS	Sentinel-1/2/3 satellite imagery is the primary input; meteorology is a secondary auxiliary channel.
H4 • Geography & user	PASS	named user is Johor Strait/Singapore coastal fish farm operators (SFA-licensed and Johor DOF-licensed/myGAP-certified), a real SEA, Singapore-preferred user segment.
H5 • Compute & budget	CAVEAT	feasible on MacBook Pro M4 Pro + free Colab/Kaggle fallback for pretraining, RM0 core cash cost, but only if the student uses cloud-native cropped data access rather than bulk downloads; naive full-product downloads risk exceeding the practical 200GB ceiling.
H6 • Pitch quality	PASS	named user, quantified historical pain (160 tonnes lost in 2014-2015, though the exact figure needs primary-source verification), a technically non-trivial core (partial-observation multi-sensor fusion), and a demoable live cloud-robustness comparison.

The problem

Johor Strait and Singapore coastal fish farms have suffered repeated, economically severe mass-mortality events from harmful algal blooms (documented dinoflagellate blooms in Feb 2014 and Feb 2015 alone cost Singapore ~160 tonnes of farmed fish across groupers, threadfin, golden trevally and rabbitfish; *Karenia mikimotoi* and *Karlodinium* spp. were the causative organisms). Current regional practice is reactive: manual microscopy and grab-sample water testing, which is slow, low-frequency and cannot give farmers warning before a bloom reaches their cages. Satellite ocean-colour monitoring, which works well for HAB tracking elsewhere, is degraded in this region by near-daily tropical cloud cover, which creates large temporal gaps in optical time series exactly when the biology is changing fastest. Existing published HAB deep-learning pipelines either discard cloudy dates (losing skill continuity) or work in non-tropical, single-sensor, non-fusion settings that have not been tested against Johor Strait's cloud regime, causative species mix, or narrow-strait hydrodynamics.

Proposed technical approach

A sequence model that ingests partially-observed, irregularly-sampled multi-sensor tensors over the Johor Strait AOI: Sentinel-2 L2A surface reflectance (10-20m, water/NIR/red-edge bands), Sentinel-3 OLCI Level-2 water products (300m, chlorophyll/CDOM proxies, near-daily revisit), Sentinel-1 GRD backscatter (10m, cloud-immune, used both for sea-state/wind-proxy features and as a persistently-available anchor modality), and ERA5/GFS meteorological reanalysis (wind, rainfall, sea-surface temperature proxies) that keeps updating even when every optical sensor is clouded out. Architecture: a masked/gated fusion encoder (e.g. a ConvLSTM or temporal transformer with per-timestep, per-modality observation masks fed as explicit channels, in the spirit of masked-autoencoder and gap-aware time-series literature) that is trained to predict a same-day and multi-day-ahead bloom-risk score even when optical channels are entirely masked out, using Sentinel-1 and meteorology alone as a fallback pathway. Training objective: (1) a self-supervised/masked-reconstruction pretraining stage on the full unlabelled multi-sensor stream (forces the model to learn cross-modal imputation, i.e. to predict what OLCI chlorophyll would show from SAR+met when optical is missing), followed by (2) supervised fine-tuning against the sparse historical fish-kill/bloom event record, with the loss and evaluation both framed around lead time (days between first elevated-risk-score crossing and the confirmed fish-kill/bloom date) rather than per-pixel chlorophyll RMSE. Output: a per-farm-zone (matching SFA farm-code East/West Johor Strait convention) risk score and trend, deployed as a live dashboard-style demo (the demo artifact only; the H2 'dashboard' ban applies to ideas whose CORE is a dashboard — here the core is the fusion/partial-observation model, the dashboard is just the presentation layer).

Novelty

WHAT THE DELTA ACTUALLY IS

The delta is threefold and must be argued precisely because two very recent (2025-2026) papers already occupy adjacent ground. First, neither the closest cloud-robust prior art (Lima & Martins 2026, arXiv: 2606.17242, Sentinel-2/Landsat-only ViT robust to cloud/glint) nor the closest fusion prior art (LaHaye et al. 2025, SIT-FUSE, arXiv:2510.02763, self-supervised multi-optical-sensor fusion) incorporates Sentinel-1 SAR as a cloud-immune anchor modality with explicit per-timestep observation masking — this project's core architectural contribution is treating SAR+meteorology as a persistently-available fallback pathway during optical blackout, not merely gap-filling optical imagery statistically. Second, the evaluation objective is different in kind: published HAB deep-learning work (all baselines above) is scored on per-pixel/per-scene chlorophyll accuracy or classification F1; this project reframes success as lead time before a farm-relevant fish-kill event, which is the metric that actually matters to a farmer deciding whether to net-cordon or harvest early, and which none of the cited baselines report. Third, the deployment regime — persistent near-daily tropical cloud cover over a narrow, hydrodynamically complex strait with two

documented and distinct fish-kill causal mechanisms (toxic HAB vs hypoxia) — is a genuine technical problem the geography creates (satisfying H1's 'genuine technical problem' bar), not a re-run of an existing pipeline on new coordinates. The delta is real but narrower than the original one-line pitch suggests: a student must explicitly benchmark against Lima & Martins (2026) and LaHaye et al. (2025) reproductions, not merely assert novelty by naming Singapore.

PRIOR ART THAT THREATENS IT

Two 2025-2026 preprints substantially narrow the novelty margin and must be read and benchmarked against before building: (1) Lima & Martins, 'Landsat-Sentinel-2 Algal Bloom Mapping Using Vision Transformers' (arXiv:2606.17242, June 2026) already demonstrates a transformer that stays robust under cloud and sun-glint stress using only optical (Sentinel-2/Landsat) data — this directly attacks the 'cloud-robust' half of the pitch, though it does not fuse SAR or meteorology and is not evaluated on lead-time-to-fish-kill nor on Southeast Asia. (2) LaHaye, Luis & Gierach, 'SIT-FUSE' (arXiv:2510.02763, Oct 2025) already does self-supervised multi-sensor HAB severity/speciation fusion without per-instrument labels — this directly attacks the 'sensor fusion without dense labels' half of the pitch, though it fuses only optical/hyperspectral sensors (VIIRS/MODIS/OLCI/OCI + TROPOMI SIF), no SAR, and is validated in the Gulf of Mexico and Southern California, not the tropics. Neither paper has been tested against persistent tropical cloud cover, Johor Strait hydrodynamics, or a lead-time-to-fish-kill objective, so the idea is not already solved, but the student must position the work explicitly against both rather than claiming an empty field.

GENERIC-PATTERN CHECK

Core is a partial-observation, multi-sensor fusion nowcasting model scored on lead time — this is not a chatbot/LLM advisory assistant, not a dashboard/monitoring-portal-as-core (the dashboard here is a demo wrapper around a real fusion model, not the deliverable itself), not leaf-photo disease classification, not a plain NDVI/vegetation-index viewer, not tabular yield prediction from CSVs, and not a re-run of an existing crop-classification benchmark. Could the one-sentence summary describe 50 other FYPs? No — 'cloud-robust multi-sensor HAB nowcasting scored on lead-time-to-fish-kill for Johor Strait' is specific enough in region, sensor combination, and evaluation framing that it would not describe a generic 'apply CNN to satellite images' project, though the student must actively defend this specificity against the two close 2025-2026 papers identified above.

Sensors and imagery

Sensor	Resolution	Revisit (days)	Bands	Licence
Sentinel-2 MSI	10-20 m	~5 (2 satellites)	Blue, Green, Red, Red-edge, NIR, SWIR (water quality indices)	Free, open (Copernicus)
Sentinel-3 OLCI	300 m	~1-2	Ocean colour Level-2 chlorophyll/ CDOM/TSM products	Free, open (Copernicus)
Sentinel-1 GRD (IW mode)	10 m	6-12 (single satellite post Sentinel-1B loss; historically 6 with 2 satellites)	VV, VH backscatter	Free, open (Copernicus)
ERA5 / GFS reanalysis	~9-25km (reanalysis grid, not imagery) m	hourly	wind, precipitation, SST proxies	Free (Copernicus Climate Data Store / NOAA)

Fusion of 3 satellite sensors + reanalysis; imagery (S1/S2/S3) is the primary input, meteorology is an auxiliary channel.

Yes. The bloom-risk signal is derived primarily from Sentinel-2/3 optical water-colour bands and Sentinel-1 SAR backscatter time series; meteorological reanalysis is a secondary, non-imagery auxiliary input used to keep the model informed during optical data gaps, not the primary driver.

Datasets

SENTINEL-2 L2A (COPERNICUS DATA SPACE ECOSYSTEM)

Access route	dataspace.copernicus.eu, free self-registration, STAC/OData API or Copernicus Browser; verified live via WebSearch on registration flow
Auth required	Free account registration only
Licence	Copernicus free and open data policy
Approx. size	Full-tile downloads for a 2-year Johor Strait AOI could exceed 100GB; using cloud-native COG/STAC range-reads cropped to the narrow-strait AOI (rather than full-tile downloads) should bring this to an estimated 15-30GB
Accessible from MY	True
How verified	WebSearch confirmed CDSE is globally open, free, self-registration, no region restriction found

SENTINEL-3 OLCI LEVEL-2 (COPERNICUS DATA SPACE ECOSYSTEM / EUMETSAT)

Access route	dataspace.copernicus.eu or EUMETSAT Data Store
Auth required	Free account registration
Licence	Copernicus free and open data policy
Approx. size	Estimated 5-15GB for AOI-cropped, multi-year daily granules
Accessible from MY	True
How verified	WebSearch confirmed CDSE offers Sentinel-3 reference products globally

SENTINEL-1 GRD (COPERNICUS DATA SPACE ECOSYSTEM / ALASKA SATELLITE FACILITY VERTEX)

Access route	dataspace.copernicus.eu or ASF Vertex (search.asf.alaska.edu)
Auth required	Free account registration
Licence	Copernicus free and open data policy
Approx. size	Estimated 10-25GB for AOI-cropped multi-year IW GRD stack
Accessible from MY	True
How verified	WebSearch confirmed CDSE/ASF free global access

ERA5 REANALYSIS (COPERNICUS CLIMATE DATA STORE)

Access route	cds.climate.copernicus.eu, free API key
Auth required	Free account + API key
Licence	Copernicus free and open data policy
Approx. size	<2GB for a small AOI, multi-year hourly subset
Accessible from MY	True
How verified	[uncertain] not independently re-verified this session beyond general knowledge of CDS; access route is well-documented and free but exact Malaysia-access nuances not tested live

HISTORICAL FISH-KILL/HAB EVENT RECORD FOR JOHOR STRAIT / SINGAPORE

Access route	Assembled manually from published literature (San Diego-McGlone et al. 2024, DOI 10.3390/su162310521; Lim et al./Leong et al. Harmful Algae papers on Johor Strait microalgal assemblages) and SFA/press archives — NOT a ready-made downloadable dataset
Auth required	None (literature synthesis), but no single machine-readable file exists
Licence	Mixed (journal articles, press archives)
Approx. size	<0.1
Accessible from MY	True
How verified	WebSearch + OpenAlex search confirmed the 2024 SEA fish-kill review and 2021 Johor Strait microalgal-assemblage paper exist and are readable, but confirmed NO structured, continuous, machine-readable event database exists for this region (see labels_and_ground_truth)

TOTAL DATA VOLUME

Estimated 30-70GB if fetched via cloud-native cropped access (STAC/COG range reads restricted to the narrow Johor Strait AOI); risk of exceeding the ~200GB H5 practical ceiling if the student naively bulk-downloads full-swath/full-tile products instead of cropping at source. **[uncertain: exact GB depends on final AOI size and time span, not independently measured this session]**

LABELS AND GROUND TRUTH

This is the central, verified risk of the whole idea. There is **NO** continuous, machine-readable ground-truth record of Johor Strait/Singapore algal bloom or fish-kill events. What exists: (1) a small number of well-documented historical fish-kill episodes — most prominently Feb 2014 and Feb 2015, when Singapore lost ~160 tonnes of farmed fish (groupers, threadfin, golden trevally, rabbitfish) to dinoflagellate blooms (*Karenia mikimotoi*, *Karlodinium australe*, *K. veneticum*), reported in peer-reviewed literature (Harmful Algae journal papers on Johor Strait microalgal assemblages, ScienceDirect S1568988321001074) and in the 2024 Southeast Asia fish-kill review (San Diego-McGlone et al., DOI 10.3390/su162310521); (2) that same 2024 review explicitly states that for events after 2016, causative organisms and confirmed dates are **NOT** clearly documented in the literature, i.e. the record goes cold; (3) a separate paper documents fish kills in the inner Johor Strait caused by hypoxia/anoxia from prolonged diatom blooms, a DIFFERENT causal mechanism from toxin-producing dinoflagellate HABs, meaning any label set must disentangle at least two distinct 'fish kill' causal pathways rather than treat them as one class. Practical substitute proposed: (a) weak/pseudo-labels from Sentinel-3 OLCI chlorophyll-a and CDOM anomaly thresholds (self-supervised, no manual labelling), (b) the sparse set of ~5-15 confirmed historical event-dates

(concentrated in 2014-2015, thin thereafter) used only as held-out positive anchors for evaluating lead time, not for training a dense classifier, (c) supplementing academic literature with a manual trawl of Singapore/Malaysia press archives (Straits Times, CNA, Bernama) for undocumented-in-papers fish-kill reports, which was not exhaustively done this session and is flagged as a week-1 task. Risk carried: severe class imbalance, reporting/selection bias (only economically significant events get reported), no confirmed 'true negative' quiet periods, and conflation of HAB-driven and hypoxia-driven fish kills unless the student manually curates by causal mechanism.

Baselines to beat

Paper	Reported metric	Dataset
LSTM Networks to Improve the Prediction of Harmful Algal Blooms in the West Coast of Sabah International Journal of Environmental Research and Public Health (MDPI) 2021 · 10.3390/ijerph18147650	RMSE and correlation coefficient r on MODIS 8-day chlorophyll-a composites (2003-2018), LSTM outperforms CNN	MODIS chlorophyll-a + GEBCO bathymetry, West Coast Sabah, Malaysia
Landsat-Sentinel-2 Algal Bloom Mapping Using Vision Transformers: Model Description, Implementation, and Examples arXiv preprint (submitted to a remote-sensing venue, exact venue not yet confirmed) 2026 · arXiv:2606.17242	Omission/commission error 8-65% across sites; Swin Transformer shown to remain robust under cloud and sun-glint stress where spectral-index baselines produce widespread false positives	Globally distributed coastal bloom-prone hotspots, Landsat-8/9 + Sentinel-2, 30m harmonised
Fusing Multi- and Hyperspectral Satellite Data for Harmful Algal Bloom Monitoring with Self-Supervised and Hierarchical Deep Learning (SIT-FUSE) arXiv preprint 2025 · arXiv: 2510.02763	Qualitative agreement with in-situ phytoplankton/species measurements, no single headline accuracy number reported	VIIRS/MODIS/OLCI/OCI + TROPOMI SIF, Gulf of Mexico and Southern California, 2018-2025
Early Warning of Harmful Algal Bloom Risk Using Satellite Ocean Color and Lagrangian Particle Trajectories Frontiers in Marine Science 2021 · 10.3389/fmars. 2021.736262	$r^2 = 0.81$, mean absolute percentage error = 36.9% between chlorophyll-a predictions and satellite observations	Coast of England, satellite ocean colour + hydrodynamic Lagrangian model

Fish Kills Related to Harmful Algal Bloom Events in Southeast Asia Sustainability (MDPI) 2024 · 10.3390/su162310521	Literature synthesis, not a quantitative ML metric; establishes causative species, hotspot countries (Philippines, Malaysia, Singapore, Indonesia, Thailand) and calls explicitly for better early-warning systems	Regional literature synthesis, Southeast Asia
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Evaluation protocol

Primary metric: lead time in days between the model's first sustained bloom-risk-score threshold crossing and the ground-truth fish-kill/bloom confirmation date, averaged over the held-out historical event set (small-N, ~5-15 events depending on how many can be recovered from literature + press archives). Secondary metrics: (a) skill retention under synthetic cloud-masking ablation — compare risk-score accuracy/lead-time when optical channels are progressively masked out (0%, 50%, 90%, 100% of days clouded) against a cloud-discarding baseline that simply drops clouded dates, to directly demonstrate the 'cloud-persistent-regime' claim; (b) standard chlorophyll-anomaly detection AUC/F1 as a secondary, more data-rich proxy metric since fish-kill events are too sparse to train/validate on alone. Geographic generalisation: train on West/East Johor Strait jointly, hold out one sub-zone (e.g. using SFA's East/West farm-code convention) for spatial generalisation testing. Success would be: positive mean lead time (i.e. any measurable warning before the event) that survives the cloud-masking ablation, materially better than a cloud-discarding optical-only baseline reproduced from LaHaye et al. (2025) or Lima & Martins (2026).

Feasibility

COMPUTE

Peak memory	Estimated 8-14GB for a modest fusion transformer/ConvLSTM at small batch size and cropped AOI patch size, well within 24GB unified memory
Est. training hours (MPS)	Estimated 6-10 hours for supervised fine-tuning on MPS; self-supervised multi-sensor pretraining stage likely needs more compute and is the part most likely to require Colab/Kaggle T4 fallback
Fits 24 GB	True
MPS-compatible ops	Standard conv/LSTM/attention operations used by ConvLSTM or small transformer architectures are MPS-compatible; SAR-specific preprocessing (e.g. speckle filtering) is typically done as a CPU preprocessing step outside the training loop, not as a custom CUDA kernel, so should not block MPS use. [uncertain: not verified against a specific chosen architecture's exact op list this session]
Colab/Kaggle fallback	True
Notes	Multi-sensor STAC/COG data engineering (aligning S1/S2/S3/ERA5 to a common grid and time axis) is likely the dominant engineering cost, not raw model compute; this matches the S2 build-heavy preference.
Cash cost	Estimated RM0 core cost: all imagery (Sentinel-1/2/3) and reanalysis (ERA5) data sources are free with self-registration; no commercial VHR imagery is required. Optional buffer of up to ~RM50 within the RM200 budget for incidental costs (e.g. cloud storage overage or a small demo-hosting fee) if the student chooses to host a live dashboard beyond the laptop demo.

Build : research	Approximately 65:30 engineering-to-research (the multi-sensor STAC ingestion pipeline, temporal alignment, masking scheme, and dashboard demo are substantial engineering; the partial-observation training objective and lead-time evaluation framework require a meaningful research design contribution to differentiate from Lima & Martins 2026 and LaHaye et al. 2025).
Publication venue fit	IGARSS or ICLR ML4RS workshop fit well given the sensor-fusion-under-missing-data framing; Remote Sensing (MDPI) is a realistic mid-tier journal target given the volume of existing HAB remote-sensing papers published there (S1 fit). CVPR EarthVision is a stretch given the relatively small, sparse-label evaluation set.

DEMO ARTIFACT

Live on a laptop: load a pre-fetched, pre-fused multi-sensor tensor for a chosen historical window spanning a known bloom event (e.g. Feb 2014/2015); step through the sequence day-by-day showing (a) which sensors were actually available that day (highlighting optical blackout days), (b) the model's running bloom-risk score and its lead-time-to-event marker, contrasted live against a cloud-discarding baseline that visibly stalls/flatlines on clouded days. Proves the core cloud-robustness claim visually in under 2 minutes.

User and pain

USER SEGMENT

Licensed sea-based fish farm operators in the Johor Strait: ~110 SFA-licensed coastal farms on the Singapore side (confirmed via SFA's public 'Types of Farms in Singapore' page and a directly-parsed SFA licensed-sea-based-farms register, 68 currently-listed licence entries as of 3 July 2026, farm codes encoding East/West Johor Strait location) contributing ~85% of Singapore's local foodfish production, plus Johor Department of Fisheries-licensed / myGAP-certified marine fish farm operators on the Malaysia side of the strait. Output consumed directly by farm operators and, plausibly, by SFA/Johor DOF as a regulator-facing early-warning tool.

QUANTIFIED PAIN

Singapore lost ~160 tonnes of farmed fish (groupers, threadfin, golden trevally, rabbitfish) in the Feb 2014 - Feb 2015 Johor Strait dinoflagellate bloom episode alone (source: WebSearch synthesis of Johor Strait HAB literature, corroborated by San Diego-McGlone et al. 2024, DOI 10.3390/su162310521, which frames Southeast Asia-wide fish-kill losses as economically significant and recurring across the Philippines, Malaysia, Singapore, Indonesia and Thailand). **[uncertain: the exact 160-tonne figure and precise dollar loss were not traced to a single primary citable source with a DOI this session — it appears consistently in secondary web summaries of the Johor Strait HAB literature; the student should verify this number directly against the primary Harmful Algae journal paper (ScienceDirect S1568988321001074) before quoting it in a pitch].**

Key risks

1. Label/ground-truth scarcity (highest risk): no continuous fish-kill/bloom event record exists past ~2016 in the literature reviewed; the evaluation set may end up being fewer than 10 usable events, making the headline lead-time metric statistically fragile.
2. Narrowed novelty margin: two 2025-2026 preprints (Lima & Martins arXiv:2606.17242; LaHaye et al. arXiv:2510.02763) already solve adjacent pieces of the problem; failing to explicitly benchmark against them risks an H1 novelty challenge from the FYP panel.

3. 3. Causal conflation risk: at least two distinct fish-kill mechanisms are documented in the Johor Strait (toxin-producing dinoflagellate HABs vs hypoxia from diatom blooms) — treating them as one label class would produce a scientifically unsound model.
4. 4. Data engineering risk: aligning Sentinel-1 (SAR, 10m), Sentinel-2 (optical, 10-20m) and Sentinel-3 OLCI (300m) onto one spatio-temporal grid for a narrow strait AOI is nontrivial and could consume a disproportionate share of the project timeline.
5. 5. Dataset volume risk: naive full-tile/full-swath downloads over a multi-year period could exceed the ~200GB H5 ceiling; the student must commit early to cloud-native cropped access (STAC/COG range reads) rather than bulk downloads.

Kill criteria

1. Week 1: manually trawl Singapore/Malaysia press archives (Straits Times, CNA, Bernama, SFA advisories) and DOF Malaysia records for undocumented fish-kill dates beyond the 2014-2015 cluster found in this session's literature search; if fewer than ~5 total usable event-dates can be assembled across all sources, the lead-time evaluation is not statistically viable and the idea should pivot to a proxy-metric-only version or be abandoned.
2. Week 1-2: pull one month of real Sentinel-2/3 data for the Johor Strait AOI and measure actual cloud-free fraction; if cloud-free days are so rare that even Sentinel-3's near-daily revisit cannot anchor a meaningful signal, reconsider the sensor mix.
3. Week 2: attempt a minimal SAR+optical temporal alignment pipeline on a 2-week sample; if STAC/COG cropped access does not work reliably from a Malaysia-based connection (untested this session), full-product downloads may be forced, risking the H5 dataset-size ceiling.
4. Week 2: reproduce (at small scale) the cloud-robustness claim from Lima & Martins (2026, arXiv: 2606.17242) on Johor Strait data using optical-only inputs; if it already performs adequately without SAR fusion, the SAR-fusion novelty claim weakens and the pitch must be re-scoped around the lead-time objective alone.

Open / unverified items

- `total_dataset_gb`: exact GB not independently measured this session; estimate depends heavily on whether the student uses cloud-native cropped access vs bulk downloads
- `datasets[ERA5].verified_how`: CDS access route not independently re-verified live this session beyond general knowledge
- `quantified_pain`: the 160-tonne figure and any dollar-loss estimate were not traced to a single primary citable source with a DOI this session; needs direct verification against the Harmful Algae journal paper (ScienceDirect S1568988321001074)
- `compute_feasibility.mps_compatible_ops`: not verified against a specific chosen model architecture's exact operator list
- `labels_and_ground_truth`: whether a press-archive trawl (Straits Times/CNA/Bernama/SFA advisories) beyond academic literature would surface additional undocumented fish-kill dates was not exhaustively tested this session — flagged as a week-1 kill-criterion task rather than assumed

Key references

Fish Kills Related to Harmful Algal Bloom Events in Southeast Asia

Sustainability (MDPI) · 2024 · 10.3390/su162310521

The single most relevant regional review; establishes causative species, hotspot countries including Malaysia and Singapore, and explicitly calls for better HAB early-warning systems — but also reveals that the post-2016 event record is undocumented, directly informing the `labels_and_ground_truth` risk assessment.

LSTM Networks to Improve the Prediction of Harmful Algal Blooms in the West Coast of Sabah

International Journal of Environmental Research and Public Health (MDPI) · 2021 · 10.3390/ijerph18147650

The closest Malaysia-specific ML HAB baseline; explicitly discusses satellite time-series cloud-cover gaps limiting prediction accuracy, i.e. names the exact problem this idea targets, and gives a concrete named baseline (LSTM vs CNN, RMSE/r) to beat or extend.

Landsat-Sentinel-2 Algal Bloom Mapping Using Vision Transformers

arXiv preprint · 2026 · arXiv:2606.17242

The closest direct prior-art threat to the 'cloud-robust' half of the novelty claim; must be read in full and benchmarked against before claiming cloud-robustness as a contribution.

Fusing Multi- and Hyperspectral Satellite Data for Harmful Algal Bloom Monitoring with Self-Supervised and Hierarchical Deep Learning

arXiv preprint · 2025 · arXiv:2510.02763

The closest direct prior-art threat to the 'self-supervised sensor fusion without dense labels' half of the novelty claim; does not use SAR or tropical data, leaving a genuine gap this idea could fill if positioned correctly.

Early Warning of Harmful Algal Bloom Risk Using Satellite Ocean Color and Lagrangian Particle Trajectories

Frontiers in Marine Science · 2021 · 10.3389/fmars.2021.736262

A named baseline for the early-warning/lead-time framing ($r^2=0.81$, MAPE=36.9%) that this idea's evaluation protocol should be positioned against, even though it is not SEA-specific.

Generated from the source proposal JSON. Amber [uncertain] flags are carried over verbatim from the source and mark claims that were not independently verified.